7

Waves in inhomogeneous plasmas and wave-energy relations

7.1 Wave propagation in inhomogeneous plasmas

Thus far in our discussion of wave propagation it has been assumed that the plasma is spatially uniform. While this assumption simplifies analysis, the real world is usually not so accommodating and it is plausible that spatial non-uniformity might modify wave propagation. The modification could be just a minor adjustment or it could be profound. Spatial non-uniformity might even produce entirely new kinds of waves. As will be seen, all these possibilities can occur.

To determine the effects of spatial non-uniformity, it is necessary to re-examine the original system of partial differential equations from which the wave dispersion relation was obtained. This is because the technique of substituting $\mathrm{i}\mathbf{k}$ for ∇ is, in essence, a shortcut for spatial Fourier analysis, and so is mathematically valid *only* if the equilibrium is spatially uniform. The criteria for whether or not ∇ can be replaced by $\mathrm{i}\mathbf{k}$ can be understood by considering the simple example of a high-frequency electromagnetic plasma wave propagating in an unmagnetized three-dimensional plasma having a gentle density gradient. The plasma frequency will be a function of position for this situation. To keep matters simple, the density non-uniformity is assumed to be in one direction only, which will be labeled the x direction. The plasma is thus uniform in the y and z directions, but non-uniform in the x direction. Because the frequency is high, ion motion may be neglected and the electron motion is just the quiver velocity

$$\mathbf{v}_1 = -\frac{q_e}{\mathrm{i}\omega m_e}\mathbf{E}_1. \tag{7.1}$$

The current density associated with electron motion is therefore

$$\mathbf{J}_1 = -\frac{n_e(x) q_e^2}{\mathrm{i}\omega m_e}\mathbf{E}_1 = -\varepsilon_0\frac{\omega_{pe}^2(x)}{\mathrm{i}\omega}\mathbf{E}_1. \tag{7.2}$$

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Inserting this current density into Ampère's law gives

$$\nabla \times \mathbf{B}_1 = -\frac{\omega_{pe}^2(x)}{\mathrm{i}\omega c^2}\mathbf{E}_1 - \frac{\mathrm{i}\omega}{c^2}\mathbf{E}_1 = -\frac{\mathrm{i}\omega}{c^2}\left(1 - \frac{\omega_{pe}^2(x)}{\omega^2}\right)\mathbf{E}_1. \tag{7.3}$$

Substituting Ampère's law into the curl of Faraday's law gives

$$\nabla \times \nabla \times \mathbf{E}_1 = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_{pe}^2(x)}{\omega^2}\right)\mathbf{E}_1. \tag{7.4}$$

Attention is now restricted to waves for which $\nabla \cdot \mathbf{E}_1 = 0$; this is a generalization of the assumption that the waves are transverse (i.e., have $\mathrm{i}\mathbf{k} \cdot \mathbf{E} = 0$) or equivalently are electromagnetic, and so involve no density perturbation. In this case, expansion of the left-hand side of Eq. (7.4) yields

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\mathbf{E}_1 + \frac{\omega^2}{c^2}\left(1 - \frac{\omega_{pe}^2(x)}{\omega^2}\right)\mathbf{E}_1 = 0. \tag{7.5}$$

It should be recalled that Fourier analysis is restricted to equations with constant coefficients, so Eq. (7.5) can only be Fourier transformed in the y and z directions. It cannot be Fourier transformed in the x direction because the coefficient $\omega_{pe}^2(x)$ depends on x. Thus, after performing only the allowed Fourier transforms, the wave equation becomes

$$\left(\frac{\partial^2}{\partial x^2} - k_y^2 - k_z^2\right)\tilde{\mathbf{E}}_1(x, k_y, k_z) + \frac{\omega^2}{c^2}\left(1 - \frac{\omega_{pe}^2(x)}{\omega^2}\right)\tilde{\mathbf{E}}_1(x, k_y, k_z) = 0, \tag{7.6}$$

where $\tilde{\mathbf{E}}_1(x, k_y, k_z)$ is the Fourier transform in the y and z directions. This may be rewritten as

$$\left(\frac{\partial^2}{\partial x^2} + \kappa^2(x)\right)\tilde{\mathbf{E}}_1(x, k_y, k_z) = 0, \tag{7.7}$$

where

$$\kappa^2(x) = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_{pe}^2(x)}{\omega^2}\right) - k_y^2 - k_z^2. \tag{7.8}$$

We now realize that Eq. (7.7) is just the spatial analog of the WKB equation for a pendulum with slowly varying frequency, namely Eq. (3.17); the only difference is that the independent variable t has been replaced by the independent variable x. Since changing the name of the independent variable is of no consequence, the solution here is formally the same as the previously derived WKB solution, Eq. (3.24). Thus the approximate solution to Eq. (7.8) is

$$\tilde{\mathbf{E}}_1(x, k_y, k_z) \sim \frac{1}{\sqrt{\kappa(x)}}\exp\left(\mathrm{i}\int^x \kappa(x')\mathrm{d}x'\right). \tag{7.9}$$

Equation (7.9) shows that both the wave amplitude and effective wavenumber vary as the wave propagates in the x direction, i.e., in the direction of the inhomogeneity. It is clear that if the inhomogeneity is in the x direction, the wavenumbers k_y and k_z do not change as the wave propagates. This is because, unlike for the x direction, it was possible to Fourier transform in the y and z directions and so k_y and k_z are just coordinates in Fourier space. The effective wavenumber in the direction of the inhomogeneity, i.e., $\kappa(x)$, is not a coordinate in Fourier space because Fourier transformation was not allowed in the x direction. The spatial dependence of the effective wavenumber $\kappa(x)$ defined by Eq. (7.8) and the spatial dependence of the WKB amplitude together provide the means by which the system accommodates the spatial inhomogeneity. The invariance of the wavenumbers in the homogeneous directions is called Snell's law. An elementary example of Snell's law is the situation where light crosses an interface between two media having different dielectric constants and the refractive index parallel to the interface remains invariant.

One way of interpreting this result is to state that the WKB method gives qualified permission to Fourier analyze in the x direction. To the extent that such an x-direction Fourier analysis is allowed, $\kappa(x)$ can be considered as the effective wavenumber in the x direction, i.e., $\kappa(x) = k_x(x)$. The results in Chapter 3 imply that the WKB approximate solution, Eq. (7.9), is valid only when the criterion

$$\frac{1}{k_x}\frac{\mathrm{d}k_x}{\mathrm{d}x} \ll k_x \tag{7.10}$$

is satisfied. Inequality (7.10) is thus not satisfied when $k_x \to 0$, i.e., at a cutoff. At a resonance the situation is somewhat more complicated. According to cold plasma theory, k_x simply diverges at a resonance; however, when hot plasma effects are taken into account, it is found that instead of having k_x going to infinity, the resonant cold plasma mode coalesces with a hot plasma mode as shown in Fig. 7.1. At the point of coalescence $\mathrm{d}k_x/\mathrm{d}x \to \infty$ while all the other

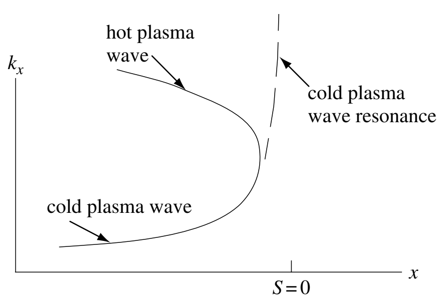

Fig. 7.1 Example of coalescence of a cold plasma wave and a hot plasma wave near the resonance of the cold plasma wave. Here a hybrid resonance causes the cold resonance.

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terms in Eq. (7.9) remain finite, and so the WKB method also breaks down at a resonance.

An interesting and important consequence of this discussion is the very real possibility that inequality (7.10) could be violated in a plasma having only the mildest of inhomogeneities. This breakdown of WKB in an apparently benign situation occurs because the critical issue is how $k_x(x)$ changes and not how plasma parameters change. For example, k_x could go through zero at some critical plasma density and, no matter how gentle the density gradient is, there will invariably be a cutoff at the critical density.

7.2 Geometric optics

The WKB method can be generalized to a plasma that is inhomogeneous in more than one dimension. In the general case of inhomogeneity in all three dimensions, the three components of the wavenumber will be functions of position, i.e., $\mathbf{k} = \mathbf{k}(\mathbf{x})$. How is the functional dependence determined? The answer is to write the dispersion relation as

$$D(\mathbf{k}, \mathbf{x}) = 0. \tag{7.11}$$

The $\mathbf{x}$-dependence of D denotes an explicit spatial dependence of the dispersion relation due to density or magnetic field gradients. This dispersion relation is now presumed to be satisfied at some initial point $\mathbf{x}$ and then it is further assumed that all quantities evolve in such a way as to keep the dispersion relation satisfied at other positions. Thus, at some arbitrary nearby position $\mathbf{x} + \delta\mathbf{x}$, the dispersion relation is also satisfied so

$$D(\mathbf{k} + \delta\mathbf{k}, \mathbf{x} + \delta\mathbf{x}) = 0 \tag{7.12}$$

or, on Taylor expanding,

$$\delta\mathbf{k} \cdot \frac{\partial D}{\partial \mathbf{k}} + \delta\mathbf{x} \cdot \frac{\partial D}{\partial \mathbf{x}} = 0. \tag{7.13}$$

The general condition for satisfying Eq. (7.13) can be established by assuming that both $\mathbf{k}$ and $\mathbf{x}$ depend on some parameter that increases monotonically along the trajectory of the wave, for example the distance s along the wave trajectory. The wave trajectory itself can also be parametrized as a function of s. Then using both $\mathbf{k} = \mathbf{k}(s)$ and $\mathbf{x} = \mathbf{x}(s)$, it is seen that moving a distance δs corresponds to respective increments $\delta\mathbf{k} = \delta s\, \mathrm{d}\mathbf{k}/\mathrm{d}s$ and $\delta\mathbf{x} = \delta s\, \mathrm{d}\mathbf{x}/\mathrm{d}s$. This means that Eq. (7.13) can be expressed as

$$\left[\frac{\mathrm{d}\mathbf{k}}{\mathrm{d}s} \cdot \frac{\partial D}{\partial \mathbf{k}} + \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}s} \cdot \frac{\partial D}{\partial \mathbf{x}}\right] \delta s = 0. \tag{7.14}$$

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The general solutions to this equation are the two coupled equations

$$\frac{\mathrm{d}\mathbf{k}}{\mathrm{d}s} = -\frac{\partial D}{\partial \mathbf{x}}, \tag{7.15}$$

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}s} = \frac{\partial D}{\partial \mathbf{k}}. \tag{7.16}$$

These are just Hamilton's equations with the dispersion relation D acting as the Hamiltonian, the path length s acting like the time, $\mathbf{x}$ acting as the position, and $\mathbf{k}$ acting as the momentum. Thus, given the initial momentum at an initial position, the wavenumber evolution and wave trajectory can be calculated using Eqs. (7.15) and (7.16) respectively. The close relationship between wavenumber and momentum fundamental to quantum mechanics is plainly evident here. Snell's law states that the wavenumber in a particular direction remains invariant if the medium is uniform in that direction; this is clearly equivalent (cf. Eq. (7.15)) to the Hamilton–Lagrange result that the canonical momentum in a particular direction is invariant if the system is uniform in that direction.

This Hamiltonian point of view provides a useful way for interpreting cutoffs and resonances. Suppose that D is the dispersion relation for a particular mode and suppose that D can be written in the form

$$D(\mathbf{k}, \mathbf{x}) = \sum_{ij} \alpha_{ij} k_i k_j + g(\omega, n(\mathbf{x}), B(\mathbf{x})) = 0. \tag{7.17}$$

If D is construed to be the Hamiltonian, then the first term in Eq. (7.17) can be identified as the "kinetic energy" while the second term can be identified as the "potential energy." As an example, consider the simple case of an electromagnetic mode in an unmagnetized plasma that has non-uniform density, so that

$$D(\mathbf{k}, \mathbf{x}) = \frac{c^2 k^2}{\omega^2} - 1 + \frac{\omega_{pe}^2(x)}{\omega^2} = 0. \tag{7.18}$$

The wave propagation can be analyzed in analogy to the problem of a particle in a potential well. Here the kinetic energy is c^2k^2/ω^2, the potential energy is $-1 + \omega_{pe}^2(x)/\omega^2$, and the total energy is zero. This is a "wave–particle" duality formally like that of quantum mechanics, since there is a correspondence between wavenumber and momentum and between energy and frequency. Cutoffs give wave reflection in analogy to the reflection of a particle in a potential well at points where the potential energy equals the total energy. As shown in Fig. 7.2(a), when the potential energy has a local minimum, waves will be trapped in the potential well associated with this minimum. Electrostatic plasma waves can also exhibit wave trapping between two reflection points; these trapped waves are called cavitons (the analysis is essentially the same; one simply replaces c^2 by $3\omega_{pe}^2\lambda_{De}^2$). The bouncing of short-wave radio waves from the ionospheric plasma

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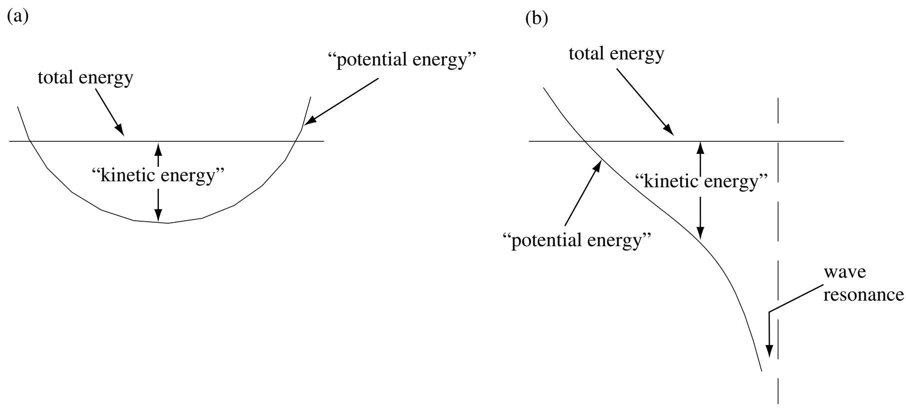

Fig. 7.2 (a) Effective potential energy for trapped wave; (b) for wave resonance.

can be analyzed using Eq. (7.18) together with Eqs. (7.15) and (7.16). As shown in Fig. 7.2(b) a wave resonance (i.e., $k^2 \to \infty$) would correspond to a deep crevice in the potential energy. One must be careful to use geometric optics only when the plasma is weakly inhomogeneous, so that the waves change sufficiently gradually to satisfy the WKB criterion.

7.3 Surface waves – the plasma-filled waveguide

An extreme form of plasma inhomogeneity occurs when there is an abrupt transition from plasma to vacuum – in other words, the plasma has an edge or surface. A qualitatively new mode, called a surface wave, appears in this circumstance. The physical basis of surface waves is closely related to the mechanism by which light waves propagate in an optical fiber.

Using the same analysis that led to Eq. (7.3), Maxwell's equations in an unmagnetized plasma may be expressed as

$$\nabla \times \mathbf{B} = -\frac{\mathrm{i}\omega}{c^2} P\mathbf{E}, \quad \nabla \times \mathbf{E} = \mathrm{i}\omega\mathbf{B}, \tag{7.19}$$

where P is the unmagnetized plasma dielectric function

$$P = 1 - \frac{\omega_{pe}^2}{\omega^2}. \tag{7.20}$$

We consider a plasma that is uniform in the z direction but non-uniform in the directions perpendicular to z. The electromagnetic fields and gradient operator can

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be separated into axial components (i.e., z direction) and transverse components (i.e., perpendicular to z) as follows:

$$\mathbf{B} = \mathbf{B}_t + B_z\hat{z}, \quad \mathbf{E} = \mathbf{E}_t + E_z\hat{z}, \quad \nabla = \nabla_t + \hat{z}\frac{\partial}{\partial z}. \tag{7.21}$$

Using these definitions Eqs. (7.19) become

$$\begin{aligned}\left(\nabla_t + \hat{z}\frac{\partial}{\partial z}\right) \times \left(\mathbf{B}_t + B_z\hat{z}\right) &= -\frac{\mathrm{i}\omega}{c^2}P\left(\mathbf{E}_t + E_z\hat{z}\right), \\ \left(\nabla_t + \hat{z}\frac{\partial}{\partial z}\right) \times \left(\mathbf{E}_t + E_z\hat{z}\right) &= \mathrm{i}\omega\left(\mathbf{B}_t + B_z\hat{z}\right).\end{aligned} \tag{7.22}$$

Since the curl of a transverse vector is in the z direction, these equations can be separated into axial and transverse components,

$$\hat{z}\cdot\nabla_t \times \mathbf{B}_t = -\frac{\mathrm{i}\omega}{c^2}PE_z, \tag{7.23}$$

$$\hat{z}\cdot\nabla_t \times \mathbf{E}_t = \mathrm{i}\omega B_z, \tag{7.24}$$

$$\hat{z}\times\frac{\partial \mathbf{B}_t}{\partial z} + \nabla_t B_z \times \hat{z} = -\frac{\mathrm{i}\omega}{c^2}P\mathbf{E}_t, \tag{7.25}$$

$$\hat{z}\times\frac{\partial \mathbf{E}_t}{\partial z} + \nabla_t E_z \times \hat{z} = \mathrm{i}\omega\mathbf{B}_t. \tag{7.26}$$

The transverse electric field on the left-hand side of Eq. (7.26) can be replaced using Eq. (7.25) to give

$$\mathrm{i}\omega\mathbf{B}_t = \hat{z}\times\frac{\partial}{\partial z}\left[\frac{\hat{z}\times\dfrac{\partial \mathbf{B}_t}{\partial z} + \nabla_t B_z \times \hat{z}}{-\dfrac{\mathrm{i}\omega}{c^2}P}\right] + \nabla_t E_z \times \hat{z}. \tag{7.27}$$

It is now assumed that all quantities have axial dependence $\sim \exp(\mathrm{i}kz)$ so Eq. (7.27) can be solved to give $\mathbf{B}_t$ solely in terms of E_z and B_z, i.e.,

$$\mathbf{B}_t = \left(\frac{\omega^2}{c^2}P - k^2\right)^{-1}\left[\nabla_t\frac{\partial B_z}{\partial z} - \frac{\mathrm{i}\omega}{c^2}P\nabla_t E_z \times \hat{z}\right]. \tag{7.28}$$

This result can also be used to solve for the transverse electric field by interchanging $-\mathrm{i}\omega P/c^2 \longleftrightarrow \mathrm{i}\omega$ and $\mathbf{E} \longleftrightarrow \mathbf{B}$ to obtain

$$\mathbf{E}_t = \left(\frac{\omega^2}{c^2}P - k^2\right)^{-1}\left[\nabla_t\frac{\partial E_z}{\partial z} + \mathrm{i}\omega\nabla_t B_z \times \hat{z}\right]. \tag{7.29}$$

Except for the plasma-dependent factor P, these are the standard waveguide equations. An important feature of these equations is that the transverse fields

$\mathbf{E}_t, \mathbf{B}_t$ are functions of the axial fields E_z, B_z only and so all that is required is to construct wave equations characterizing the axial fields. This is an enormous simplification because, instead of having to derive and solve six wave equations in the six components of $\mathbf{E}, \mathbf{B}$ as might be expected, it suffices to derive and solve wave equations for just E_z and B_z.

The sought-after wave equations are determined by eliminating $\mathbf{E}_t$ and $\mathbf{B}_t$ from Eqs. (7.23) and (7.24) to obtain

$$\hat{z} \cdot \nabla_t \times \left\{ \left(\frac{\omega^2}{c^2} P - k^2 \right)^{-1} \left[\mathrm{i}k \nabla_t B_z - \frac{\mathrm{i}\omega}{c^2} P \nabla_t E_z \times \hat{z} \right] \right\} = -\frac{\mathrm{i}\omega}{c^2} P E_z, \quad (7.30)$$

$$\hat{z} \cdot \nabla_t \times \left\{ \left(\frac{\omega^2}{c^2} P - k^2 \right)^{-1} \left[\mathrm{i}k \nabla_t E_z + \mathrm{i}\omega \nabla_t B_z \times \hat{z} \right] \right\} = \mathrm{i}\omega B_z. \quad (7.31)$$

In the special situation where $\nabla_t P \times \nabla_t B_z = \nabla_t P \times \nabla_t E_z = 0$, the first terms in the square brackets of the above equations vanish. This simplification would occur, for example, in an azimuthally symmetric plasma having an azimuthally symmetric perturbation so that $\nabla_t P$, $\nabla_t E_z$, and $\nabla_t B_z$ are all in the r direction. It is now assumed that both the plasma and the mode have this azimuthal symmetry so Eqs. (7.30) and (7.31) reduce to

$$\hat{z} \cdot \nabla_t \times \left\{ \left(\frac{\omega^2}{c^2} P - k^2 \right)^{-1} \left(P \nabla_t E_z \times \hat{z} \right) \right\} = P E_z, \quad (7.32)$$

$$\hat{z} \cdot \nabla_t \times \left\{ \left(\frac{\omega^2}{c^2} P - k^2 \right)^{-1} \left(\nabla_t B_z \times \hat{z} \right) \right\} = B_z \quad (7.33)$$

or, equivalently,

$$\nabla_t \cdot \left\{ \frac{P}{P - k^2 c^2/\omega^2} \nabla_t E_z \right\} + \frac{\omega^2}{c^2} P E_z = 0, \quad (7.34)$$

$$\nabla_t \cdot \left\{ \frac{1}{P - k^2 c^2/\omega^2} \nabla_t B_z \right\} + \frac{\omega^2}{c^2} B_z = 0. \quad (7.35)$$

The assumption that both the plasma and the modes are azimuthally symmetric has the important consequence of decoupling the E_z and B_z modes so there are two distinct polarizations. These are (i) a mode where B_z is finite but $E_z = 0$ and (ii) the reverse. Case (i) is called a transverse electric (TE) mode while case (ii) is called a transverse magnetic mode (TM), since in the first case the electric field is purely transverse while in the second case the magnetic field is purely transverse.

We now consider an azimuthally symmetric TM mode propagating in a uniform cylindrical plasma of radius a surrounded by vacuum. Since $B_z = 0$ for a TM mode, the transverse fields are the following functions of E_z:

$$\mathbf{B}_t = \left(\frac{\omega^2}{c^2}P - k^2\right)^{-1}\left[-\frac{\mathrm{i}\omega}{c^2}P\nabla_t E_z \times \hat{z}\right], \tag{7.36}$$

$$\mathbf{E}_t = \left(\frac{\omega^2}{c^2}P - k^2\right)^{-1}\nabla_t \frac{\partial E_z}{\partial z}. \tag{7.37}$$

Additionally, because of the assumed symmetry, the TM mode Eq. (7.34) simplifies to

$$\frac{1}{r}\frac{\partial}{\partial r}\left(\frac{rP}{P - k^2c^2/\omega^2}\frac{\partial E_z}{\partial r}\right) + \frac{\omega^2}{c^2}PE_z = 0. \tag{7.38}$$

Since P is uniform within the plasma region and within the vacuum region, but has different values in these two regions, separate solutions to Eq. (7.38) must be obtained in the plasma and vacuum regions respectively and then matched at the interface. The jump in P is accommodated by defining distinct radial wave numbers

$$\kappa_p^2 = k^2 - \frac{\omega^2}{c^2}P, \tag{7.39}$$

$$\kappa_v^2 = k^2 - \frac{\omega^2}{c^2} \tag{7.40}$$

for the respective plasma and vacuum regions. The solutions to Eq. (7.38) in the respective plasma and vacuum regions are linear combinations of Bessel functions of order zero. If both of κ_p^2 and κ_v^2 are positive then the TM mode has the peculiar property of being radially evanescent in *both* the plasma and vacuum regions. In this case both the vacuum and plasma region solutions consist of modified Bessel functions I_0, K_0. These solutions are constrained by physical considerations as follows:

1. Because the parallel electric field is a physical quantity it must be finite. In particular, E_z must be finite at $r = 0$, in which case only the $I_0(\kappa_p r)$ solution is allowed in the plasma region (the K_0 solution diverges at $r = 0$). Similarly, because E_z must be finite as $r \to \infty$, only the $K_0(\kappa_v r)$ solution is allowed in the vacuum region (the $I_0(\kappa_v r)$ solution diverges at $r = \infty$).
2. The parallel electric field E_z must be continuous across the vacuum–plasma interface. This constraint is imposed by Faraday's law and can be seen by integrating Faraday's law over an area in the $r - z$ plane of axial length L and infinitesimal radial width.

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The inner radius of this area is at r_- and the outer radius is at r_+, where $r_- < a < r_+$. Integrating Faraday's law over this area gives

$$\lim_{r_- \to r_+} \int \mathrm{d}\mathbf{s} \cdot \nabla \times \mathbf{E} = \oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\lim_{r_- \to r_+} \int \mathrm{d}\mathbf{s} \cdot \frac{\partial \mathbf{B}}{\partial t}$$

or

$$E_z^{vac} L - E_z^{plasma} L = 0$$

showing E_z must be continuous at the plasma–vacuum interface.

3. Integration of Eq. (7.38) across the interface shows that the Quantity

$$P\left(P - k^2c^2/\omega^2\right)^{-1} \partial E_z/\partial r$$

must be continuous across the interface.

In order to satisfy Constraint #1 the parallel electric field in the plasma must be

$$E_z(r) = E_z(a)\frac{I_0(\kappa_p r)}{I_0(\kappa_p a)} \tag{7.41}$$

and the parallel electric field in the vacuum must be

$$E_z(r) = E_z(a)\frac{K_0(\kappa_v r)}{K_0(\kappa_v a)}. \tag{7.42}$$

The normalization has been set so that E_z is continuous across the interface as required by Constraint #2.

Constraint #3 gives

$$\left[\left(\frac{\omega^2}{c^2}P - k^2\right)^{-1} P\frac{\partial E_z}{\partial r}\right]_{r=a_-} = \left[\left(\frac{\omega^2}{c^2} - k^2\right)^{-1} \frac{\partial E_z}{\partial r}\right]_{r=a_+}. \tag{7.43}$$

Inserting Eqs. (7.41) and (7.42) into the respective left- and right-hand sides of the above expression gives

$$\left(\frac{\omega^2}{c^2}P - k^2\right)^{-1} P\frac{\kappa_p I_0'(\kappa_p a)}{I_0(\kappa_p a)} = \left(\frac{\omega^2}{c^2} - k^2\right)^{-1} \frac{\kappa_v K_0'(\kappa_v a)}{K_0(\kappa_v a)}, \tag{7.44}$$

where a prime means a derivative with respect to the argument of the function. This expression is effectively a dispersion relation, since it prescribes a functional relationship between ω and k. It is qualitatively different from the previously discussed uniform plasma dispersion relations because it depends on a specific physical dimension, namely the plasma radius a. This dependence indicates that a plasma–vacuum interface is a necessary condition for the mode to exist. The mode amplitude is strongest in the vicinity of the interface because both the plasma and vacuum fields decay exponentially with increasing distance from the interface.

This surface wave dispersion depends on a combination of Bessel functions and the parallel dielectric P. However, a limit exists where the dispersion relation reduces to a simpler form, and this limit illustrates important features of these surface waves. Specifically, if the axial wavelength is sufficiently short for k^2 to be much larger than both $\omega^2 P/c^2$ and ω^2/c^2, one can then approximate $k^2 \simeq \kappa_v^2 \simeq \kappa_p^2$, in which case the dispersion simplifies to

$$P\frac{I_0'(ka)}{I_0(ka)} \simeq \frac{K_0'(ka)}{K_0(ka)}. \tag{7.45}$$

If, in addition, the axial wavelength is sufficiently long to satisfy $ka \ll 1$, then the small-argument limits of the modified Bessel functions can be used, namely

$$\lim_{\xi \ll 1} I_0(\xi) = 1 + \frac{\xi^2}{4}, \tag{7.46}$$

$$\lim_{\xi \ll 1} K_0(\xi) = -\ln \xi. \tag{7.47}$$

Thus, in the limit $\omega^2 P/c^2$, $\omega^2/c^2 \ll k^2 \ll 1/a^2$, Eq. (7.45) simplifies to

$$\left(1 - \frac{\omega_{pe}^2}{\omega^2}\right)\frac{ka}{2} \simeq -\frac{1}{ka \ln\left(\dfrac{1}{ka}\right)}. \tag{7.48}$$

Because $ka \ll 1$, the logarithmic term is negative. Hence, to satisfy Eq. (7.48) it is necessary to have $\omega \ll \omega_{pe}$ so that the dispersion further becomes

$$\frac{\omega}{\omega_{pe}} = ka\sqrt{\frac{1}{2}\ln\left(\frac{1}{ka}\right)}. \tag{7.49}$$

On the other hand, if $ka \gg 1$, then the large-argument limit of the Bessel functions can be used, namely

$$\lim_{\xi \gg 1} I_0(\xi) = \frac{1}{\sqrt{2\pi\xi}}e^{\xi}, \quad \lim_{\xi \gg 1} K_0(\xi) = \sqrt{\frac{\pi}{2\xi}}e^{-\xi} \tag{7.50}$$

so that the dispersion relation becomes

$$1 - \frac{\omega_{pe}^2}{\omega^2} = -1 \tag{7.51}$$

or

$$\omega = \frac{\omega_{pe}}{\sqrt{2}}. \tag{7.52}$$

This provides the curious result that a finite-radius plasma cylinder resonates at a lower frequency than a uniform plasma if the axial wavelength is much shorter than the cylinder radius.

These surface waves are slow waves since $\omega/k \ll c$ has been assumed. They were first studied by Trivelpiece and Gould (1959) and are seen in cylindrical

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plasmas surrounded by vacuum. For $ka \ll 1$ the phase velocity is $\omega/k \sim \mathcal{O}(\omega_{pe}a)$ and for $ka \gg 1$ the phase velocity goes to zero since ω is a constant at large ka. More complicated variations of the surface wave dispersion are obtained if the vacuum region is of finite extent and is surrounded by a conducting wall, i.e., if there is plasma for $r < a$, vacuum for $a < r < b$, and a conducting wall at $r = b$. In this case the vacuum region solution consists of a linear combination of $I_0(\kappa_v r)$ and $K_0(\kappa_v r)$ terms with coefficients chosen to satisfy Constraints #2 and #3 discussed earlier and also a new, additional constraint that E_z must vanish at the wall, i.e., at $r = b$.

7.4 Plasma wave-energy equation

The energy associated with a plasma wave is related in a subtle way to the dispersive properties of the wave. Quantifying this relation requires starting from first principles regarding the electromagnetic field energy density and taking into account specific features of dispersive waves. The basic equation characterizing electromagnetic energy density, called Poynting's theorem, is obtained by subtracting **B** dotted with Faraday's law from **E** dotted with Ampère's law,

$$\mathbf{E}\cdot\nabla\times\mathbf{B} - \mathbf{B}\cdot\nabla\times\mathbf{E} = \varepsilon_0\mu_0\mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t} + \mathbf{B}\cdot\frac{\partial\mathbf{B}}{\partial t} + \mu_0\mathbf{E}\cdot\mathbf{J},$$

and expressing this result as a conservation equation,

$$\frac{\partial w}{\partial t} + \nabla\cdot\mathbf{P} = 0. \tag{7.53}$$

The quantity

$$\mathbf{P} = \frac{\mathbf{E}\times\mathbf{B}}{\mu_0} \tag{7.54}$$

is called the Poynting flux and is the electromagnetic energy flux into the system while

$$\frac{\partial w}{\partial t} = \varepsilon_0\mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t} + \frac{1}{\mu_0}\mathbf{B}\cdot\frac{\partial\mathbf{B}}{\partial t} + \mathbf{E}\cdot\mathbf{J} \tag{7.55}$$

is the rate of change of energy density in the system. The energy density is obtained by time integration and is

$$\begin{aligned} w(t) &= w(t_0) + \int_{t_0}^{t}\mathrm{d}t\left\{\varepsilon_0\mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t} + \frac{1}{\mu_0}\mathbf{B}\cdot\frac{\partial\mathbf{B}}{\partial t} + \mathbf{E}\cdot\mathbf{J}\right\} \\ &= w(t_0) + \left[\frac{\varepsilon_0}{2}E^2 + \frac{B^2}{2\mu_0}\right]_{t_0}^{t} + \int_{t_0}^{t}\mathrm{d}t\,\mathbf{E}\cdot\mathbf{J}, \end{aligned} \tag{7.56}$$

where $w(t_0)$ is the energy density at some reference time t_0.

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The quantity $\mathbf{E}\cdot\mathbf{J}$ is the rate of change of kinetic energy density of the particles. This can be seen by first dotting the Lorentz equation with $\mathbf{v}$ to obtain

$$m\mathbf{v}\cdot\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}=q\mathbf{v}\cdot(\mathbf{E}+\mathbf{v}\times\mathbf{B}) \tag{7.57}$$

or

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}mv^2\right)=q\mathbf{E}\cdot\mathbf{v}. \tag{7.58}$$

Since this is the rate of change of kinetic energy of a single particle, the rate of change of the kinetic energy density of all the particles, found by summing over all the particles, is

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\,(\text{kinetic energy density}) &= \sum_\sigma \int \mathrm{d}\mathbf{v} f_\sigma q_\sigma \mathbf{E}\cdot\mathbf{v} \\ &= \sum_\sigma n_\sigma q_\sigma \mathbf{E}\cdot\mathbf{u}_\sigma \\ &= \mathbf{E}\cdot\mathbf{J}. \end{aligned} \tag{7.59}$$

This shows that positive $\mathbf{E}\cdot\mathbf{J}$ corresponds to work going into the particles (increase of particle kinetic energy) whereas negative $\mathbf{E}\cdot\mathbf{J}$ corresponds to work coming out of the particles (decrease of the particle kinetic energy). The latter situation is obviously possible only if the particles start with a finite initial kinetic energy. Since $\mathbf{E}\cdot\mathbf{J}$ accounts for changes in the particle kinetic energy density, w must be the sum of the electromagnetic field density and the particle kinetic energy density.

The time integration of Eq. (7.55) must be done with great care when $\mathbf{E}$ and $\mathbf{J}$ are wave fields. This is because it must always be kept in mind that writing a wave field as $\psi=\tilde{\psi}\exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x}-\mathrm{i}\omega t)$ is simply a notational convenience and should never be taken to mean that the actual physical wave field is complex. The physically meaningful variable is, of course, $\psi=\mathrm{Re}\left[\tilde{\psi}\exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x}-\mathrm{i}\omega t)\right]$. Explicitly taking the real part is critical when evaluating nonlinear relationships because, unlike for linear relationships, omission of this step would lead to a nonsensical result. When dealing with a product of two oscillating physical quantities, say $\psi(t)=\mathrm{Re}\left[\tilde{\psi}\mathrm{e}^{-\mathrm{i}\omega t}\right]$ and $\chi(t)=\mathrm{Re}\left[\tilde{\chi}\mathrm{e}^{-\mathrm{i}\omega t}\right]$, it is therefore essential to extract the real part *before* calculating the product and so the product must be written as

$$\psi(t)\chi(t)=\mathrm{Re}\left[\tilde{\psi}\mathrm{e}^{-\mathrm{i}\omega t}\right]\times\mathrm{Re}\left[\tilde{\chi}\mathrm{e}^{-\mathrm{i}\omega t}\right]. \tag{7.60}$$

In particular, if $\omega = \omega_r + i\omega_i$ is a complex frequency, then the product in Eq. (7.60) assumes the form

$$\begin{aligned}\psi(t)\chi(t) &= \left(\frac{\tilde{\psi}e^{-i\omega t} + \tilde{\psi}^* e^{i\omega^* t}}{2}\right)\left(\frac{\tilde{\chi}e^{-i\omega t} + \tilde{\chi}^* e^{i\omega^* t}}{2}\right) \\ &= \frac{1}{4}\left[\begin{array}{l}\tilde{\psi}\tilde{\chi}e^{-2i\omega t} + \tilde{\psi}^*\tilde{\chi}^* e^{2i\omega^* t} \\ +\tilde{\psi}\tilde{\chi}^* e^{-i(\omega-\omega^*)t} + \tilde{\psi}^*\tilde{\chi}e^{-i(\omega-\omega^*)t}\end{array}\right]. \end{aligned} \tag{7.61}$$

When considering the energy density of a wave, we are typically interested in time-averaged quantities, not rapidly fluctuating quantities. Thus the time average of the product $\psi(t)\chi(t)$ over one wave period will be considered. The $\tilde{\psi}\tilde{\chi}$ and $\tilde{\psi}^*\tilde{\chi}^*$ terms oscillate at the fast second harmonic of the frequency and vanish upon time-averaging. In contrast, the $\tilde{\psi}\tilde{\chi}^*$ and $\tilde{\psi}^*\tilde{\chi}$ terms survive time-averaging because these terms have no oscillatory factor since $\omega - \omega^* = 2i\omega_i$. Thus, the time average, denoted by $\langle\rangle$, of the product is

$$\begin{aligned}\langle\psi(t)\chi(t)\rangle &= \frac{1}{4}(\tilde{\psi}\tilde{\chi}^* + \tilde{\psi}^{\,*}\tilde{\chi})e^{2\omega_i t}; \\ &= \frac{1}{2}\,\mathrm{Re}\left(\tilde{\psi}\;\tilde{\chi}^*\right)e^{2\omega_i t};\end{aligned} \tag{7.62}$$

this is the correct rule for time-averaging products of oscillating physical quantities, which have been represented using complex notation.

7.5 Cold plasma wave-energy equation

The current density $\mathbf{J}$ in Ampère's law consists of the explicit plasma currents. The frequency dependence of these currents means that care is required when integrating $\mathbf{E}\cdot\mathbf{J}$. In order to prepare for this integration we express Eq. (6.9) and Eq. (6.10) as

$$\begin{aligned}\frac{1}{\mu_0}\nabla\times\tilde{\mathbf{B}} &= \varepsilon_0\frac{\partial}{\partial t}\left(\overleftrightarrow{\mathbf{K}}(\omega)\cdot\tilde{\mathbf{E}}e^{-i\omega t}\right) \\ &= \tilde{\mathbf{J}}e^{-i\omega t} + \varepsilon_0\frac{\partial}{\partial t}\left(\tilde{\mathbf{E}}e^{-i\omega t}\right),\end{aligned} \tag{7.63}$$

where the time dependence is shown explicitly and $\tilde{\mathbf{J}}$ refers to the explicit plasma currents as distinct from the displacement current.

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Integration of Eq. (7.55) taking into account the prescription given by Eq. (7.60) has the form

$$w(t) = w(-\infty) + \int_{-\infty}^{t} \mathrm{d}t \left\langle \begin{array}{c} \mathbf{E}(\mathbf{x}, t) \cdot \left(\mathbf{J}(\mathbf{x},\mathbf{t}) + \varepsilon_0 \dfrac{\partial \mathbf{E}(\mathbf{x}, t)}{\partial t} \right) \\ + \dfrac{1}{\mu_0} \mathbf{B}(\mathbf{x}, t) \cdot \dfrac{\partial \mathbf{B}(\mathbf{x}, t)}{\partial t} \end{array} \right\rangle$$

$$= w(-\infty) + \frac{1}{4} \int_{-\infty}^{t} \mathrm{d}t \left\langle \begin{array}{l} \tilde{\mathbf{E}} \mathrm{e}^{-\mathrm{i}\omega t} \cdot \varepsilon_0 \dfrac{\partial}{\partial t} \left(\overleftrightarrow{\mathbf{K}}(\omega) \cdot \tilde{\mathbf{E}} \mathrm{e}^{-\mathrm{i}\omega t} \right)^* \\ + \dfrac{1}{\mu_0} \tilde{\mathbf{B}} \mathrm{e}^{-\mathrm{i}\omega t} \cdot \dfrac{\partial}{\partial t} \left(\tilde{\mathbf{B}} \mathrm{e}^{-\mathrm{i}\omega t} \right)^* + \text{c.c.} \end{array} \right\rangle, \tag{7.64}$$

where c.c. means complex conjugate of all preceding terms in the expression. The term containing the rates of change of the electric field and the particle kinetic energy can be written using Eq. (7.63) as

$$\begin{aligned} \left\langle \mathbf{E} \cdot \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \right\rangle &= \frac{\varepsilon_0}{4} \left\{ \tilde{\mathbf{E}} \mathrm{e}^{-\mathrm{i}\omega t} \cdot \left(\mathrm{i}\omega^* \overleftrightarrow{\mathbf{K}}^* \cdot \tilde{\mathbf{E}}^* \mathrm{e}^{\mathrm{i}\omega^* t} \right) + \text{c.c.} \right\} \\ &= \frac{\varepsilon_0}{4} \left\{ \tilde{\mathbf{E}} \cdot \mathrm{i}\omega^* \overleftrightarrow{\mathbf{K}}^* \cdot \tilde{\mathbf{E}}^* - \tilde{\mathbf{E}}^* \cdot \mathrm{i}\omega \overleftrightarrow{\mathbf{K}} \cdot \tilde{\mathbf{E}} \right\} \mathrm{e}^{2\omega_i t} \\ &= \frac{\varepsilon_0}{4} \left\{ \begin{array}{l} \mathrm{i}\omega_r \left[\tilde{\mathbf{E}} \cdot \overleftrightarrow{\mathbf{K}}^* \cdot \tilde{\mathbf{E}}^* - \tilde{\mathbf{E}}^* \cdot \overleftrightarrow{\mathbf{K}} \cdot \tilde{\mathbf{E}} \right] \\ + \omega_i \left[\tilde{\mathbf{E}} \cdot \overleftrightarrow{\mathbf{K}}^* \cdot \tilde{\mathbf{E}}^* + \tilde{\mathbf{E}}^* \cdot \overleftrightarrow{\mathbf{K}} \cdot \tilde{\mathbf{E}} \right] \end{array} \right\} \mathrm{e}^{2\omega_i t}. \end{aligned} \tag{7.65}$$

To proceed further, it is noted that

$$\tilde{\mathbf{E}} \cdot \overleftrightarrow{\mathbf{K}}^* \tilde{\mathbf{E}}^* = \sum_{pq} \tilde{E}_{p} K_{pq}^* \tilde{E}_{q}^{*} = \sum_{pq} \tilde{E}_{p} K_{qp}^{*t} \tilde{E}_{q}^{*} = \tilde{\mathbf{E}}^* \cdot \overleftrightarrow{\mathbf{K}}^{\dagger} \cdot \tilde{\mathbf{E}}, \tag{7.66}$$

where the superscript t means transpose and the superscript † means Hermitian conjugate, i.e., the complex conjugate of the transpose. Thus, Eq. (7.65) can be rewritten as

$$\left\langle \mathbf{E} \cdot \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \right\rangle = \frac{\varepsilon_0}{4} \left[\mathrm{i}\omega_r \tilde{\mathbf{E}}^* \cdot \left(\overleftrightarrow{\mathbf{K}}^{\dagger} - \overleftrightarrow{\mathbf{K}} \right) \cdot \tilde{\mathbf{E}} + \omega_i \tilde{\mathbf{E}}^* \cdot \left(\overleftrightarrow{\mathbf{K}}^{\dagger} + \overleftrightarrow{\mathbf{K}} \right) \cdot \tilde{\mathbf{E}} \right] \mathrm{e}^{2\omega_i t}. \tag{7.67}$$

Both the Hermitian part of the dielectric tensor,

$$\overleftrightarrow{\mathbf{K}}_h = \frac{1}{2} \left(\overleftrightarrow{\mathbf{K}} + \overleftrightarrow{\mathbf{K}}^{\dagger} \right), \tag{7.68}$$

and the anti-Hermitian part,

$$\overleftrightarrow{\mathbf{K}}_{ah} = \frac{1}{2} \left(\overleftrightarrow{\mathbf{K}} - \overleftrightarrow{\mathbf{K}}^{\dagger} \right), \tag{7.69}$$

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occur in Eq. (7.67). The cold plasma dielectric tensor is a function of ω via the functions S, P, and D,

$$\overleftrightarrow{\mathbf{K}}(\omega)=\begin{bmatrix} S(\omega) & -\mathrm{i}D(\omega) & 0 \\ \mathrm{i}D(\omega) & S(\omega) & 0 \\ 0 & 0 & P(\omega) \end{bmatrix}. \tag{7.70}$$

If $\omega_i = 0$ then S, P, and D are all pure real. In this case $\overleftrightarrow{\mathbf{K}}(\omega)$ is Hermitian so that $\overleftrightarrow{\mathbf{K}}_h = \overleftrightarrow{\mathbf{K}}$ and $\overleftrightarrow{\mathbf{K}}_{ah} = 0$. However, if ω_i is finite but small, then $\overleftrightarrow{\mathbf{K}}(\omega)$ will have a small non-Hermitian part. This non-Hermitian part is extracted using a Taylor expansion in terms of ω_i, i.e.,

$$\overleftrightarrow{\mathbf{K}}(\omega_r+\mathrm{i}\omega_i) = \overleftrightarrow{\mathbf{K}}(\omega_r)+\mathrm{i}\omega_i\left[\frac{\partial}{\partial\omega}\overleftrightarrow{\mathbf{K}}(\omega)\right]_{\omega=\omega_r}. \tag{7.71}$$

The transpose of the complex conjugate of this expansion is

$$\left[\overleftrightarrow{\mathbf{K}}(\omega_r+\mathrm{i}\omega_i)\right]^{\dagger} = \overleftrightarrow{\mathbf{K}}(\omega_r)-\mathrm{i}\omega_i\left[\frac{\partial}{\partial\omega}\overleftrightarrow{\mathbf{K}}(\omega)\right]_{\omega=\omega_r}, \tag{7.72}$$

since $\overleftrightarrow{\mathbf{K}}$ is non-Hermitian only to the extent that ω_i is finite. Substituting Eqs. (7.71) and (7.72) into Eqs. (7.68) and (7.69) and assuming small ω_i gives

$$\overleftrightarrow{\mathbf{K}}_h = \overleftrightarrow{\mathbf{K}}(\omega_r) \tag{7.73}$$

and

$$\overleftrightarrow{\mathbf{K}}_{ah} = \mathrm{i}\omega_i\left[\frac{\partial}{\partial\omega}\overleftrightarrow{\mathbf{K}}(\omega)\right]_{\omega=\omega_r}. \tag{7.74}$$

Inserting Eqs. (7.73) and (7.74) in Eq. (7.67) yields

$$\begin{aligned}\left\langle \mathbf{E}\cdot\left(\mathbf{J}+\varepsilon_0\frac{\partial\mathbf{E}}{\partial t}\right)\right\rangle &= \frac{2\varepsilon_0\omega_i}{4}\left[\omega_r\tilde{\mathbf{E}}^*\cdot\left[\frac{\partial}{\partial\omega}\overleftrightarrow{\mathbf{K}}(\omega)\right]_{\omega=\omega_r}\cdot\tilde{\mathbf{E}}+\tilde{\mathbf{E}}^*\cdot\overleftrightarrow{\mathbf{K}}(\omega_r)\cdot\tilde{\mathbf{E}}\right]\mathrm{e}^{2\omega_i t} \\ &= \frac{2\varepsilon_0\omega_i}{4}\tilde{\mathbf{E}}^*\cdot\left[\frac{\partial}{\partial\omega}\omega\overleftrightarrow{\mathbf{K}}(\omega)\right]_{\omega=\omega_r}\cdot\tilde{\mathbf{E}}\mathrm{e}^{2\omega_i t}. \end{aligned}\tag{7.75}$$

Similarly, the rate of change of the magnetic energy density is

$$\left\langle\frac{1}{\mu_0}\mathbf{B}\cdot\frac{\partial\mathbf{B}}{\partial t}\right\rangle = \frac{1}{4\mu_0}\left[2\omega_i\tilde{\mathbf{B}}^*\cdot\tilde{\mathbf{B}}\right]\mathrm{e}^{2\omega_i t}. \tag{7.76}$$

Using Eqs. (7.75) and (7.76) in Eq. (7.64) gives

$$w = w(-\infty) + \left\{ \frac{\varepsilon_0}{4}\tilde{\mathbf{E}}^* \cdot \left[\frac{\partial}{\partial\omega}\left(\omega \overleftrightarrow{\mathbf{K}}(\omega)\right)\right]_{\omega=\omega_r} \cdot \tilde{\mathbf{E}} + \frac{1}{4\mu_0}\left[\tilde{\mathbf{B}}^* \cdot \tilde{\mathbf{B}}\right] \right\} \int_{-\infty}^{t} \mathrm{d}t 2\omega_i \mathrm{e}^{2\omega_i t}, \tag{7.77}$$

which now may be integrated in time to give the total energy density *associated with bringing the wave into existence*

$$\begin{aligned} \bar{w} &= w - w(-\infty) \\ &= \left\{ \frac{\varepsilon_0}{4}\tilde{\mathbf{E}}^* \cdot \left[\frac{\partial}{\partial\omega}\left(\omega \overleftrightarrow{\mathbf{K}}(\omega)\right)\right]_{\omega=\omega_r} \cdot \tilde{\mathbf{E}} + \frac{1}{4\mu_0}\left[\tilde{\mathbf{B}}^* \cdot \tilde{\mathbf{B}}\right] \right\} \mathrm{e}^{2\omega_i t}. \end{aligned} \tag{7.78}$$

In the limit $\omega_i \to 0$ this reduces to

$$\bar{w} = \frac{\varepsilon_0}{4}\tilde{\mathbf{E}}^* \cdot \frac{\partial}{\partial\omega}\left[\omega \overleftrightarrow{\mathbf{K}}(\omega)\right] \cdot \tilde{\mathbf{E}} + \frac{|\tilde{B}|^2}{4\mu_0}. \tag{7.79}$$

Since the time-average of the energy density stored in the oscillating vacuum electric field is

$$w_E = \frac{\varepsilon_0 |\tilde{E}\ |^2}{4} \tag{7.80}$$

and the time-average of the energy density stored in the oscillating vacuum magnetic field is

$$w_B = \frac{|\tilde{B}|^2}{4\mu_0}, \tag{7.81}$$

the change in the time-average of the particle kinetic energy density associated with bringing the wave into existence is

$$\bar{w}_{part} = \frac{\varepsilon_0}{4}\tilde{\mathbf{E}}^* \cdot \left[\frac{\partial}{\partial\omega}\left(\omega \overleftrightarrow{\mathbf{K}}(\omega)\right) - \overleftrightarrow{\mathbf{I}} \right] \cdot \tilde{\mathbf{E}}. \tag{7.82}$$

This result has been established for the general case of the dielectric tensor $\overleftrightarrow{\mathbf{K}}(\omega)$ of a cold magnetized plasma. In order to illustrate the application of Eq. (7.79), we consider the simple example of high-frequency electrostatic oscillations in an unmagnetized plasma. In this situation $S = P = 1 - \omega_{pe}^2/\omega^2$ and $D = 0$

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so $\overleftrightarrow{\mathbf{K}}(\omega) = \left(1 - \omega_{pe}^2/\omega^2\right) \overleftrightarrow{\mathbf{I}}$. Since the oscillations are electrostatic, $w_B = 0$. The energy density of the particles is therefore

$$\begin{aligned}\bar{w}_{part} &= \frac{\varepsilon_0 |\tilde{E}\ |^2}{4} \left[\frac{\partial}{\partial \omega} \left\{ \omega \left(1 - \frac{\omega_{pe}^2}{\omega^2} \right) \right\} - 1 \right] \\ &= \frac{\varepsilon_0 |\tilde{E}\ |^2}{4} \left[2 \frac{\omega_{pe}^2}{\omega^2} - 1 \right] \\ &= \frac{\varepsilon_0 |\tilde{E}\ |^2}{4},\end{aligned} \tag{7.83}$$

where the dispersion relation $1 - \omega_{pe}^2/\omega^2 = 0$ has been used. Thus, in this simple example half of the average wave-energy density is contained in the electric field while the other half is contained in the coherent particle motion associated with the wave.

7.6 Finite-temperature plasma wave-energy equation

The dielectric tensor has no dependence on the wavevector $\mathbf{k}$ in a cold plasma, but does have such a dependence in a finite-temperature plasma. For example, the electrostatic unmagnetized cold plasma dielectric $P(\omega) = 1 - \omega_{pe}^2/\omega^2$ becomes $P(\omega, k) = 1 - (1 + 3k^2\lambda_{De}^2)\omega_{pe}^2/\omega^2$ in a warm plasma. The analysis of the previous section will now be extended to allow for the possibility that the dielectric tensor depends on $\mathbf{k}$ as well as on ω. In analogy to the method used in the previous section for treating complex ω, here $\mathbf{k}$ will also be assumed to have a small imaginary part. In this case, Taylor expansion of the dielectric tensor followed by extraction of the anti-Hermitian part shows that the anti-Hermitian part is

$$\overleftrightarrow{\mathbf{K}}_{ah} = \mathrm{i}\omega_i \left[\frac{\partial}{\partial \omega} \overleftrightarrow{\mathbf{K}}(\omega, \mathbf{k}) \right]_{\omega=\omega_r, \mathbf{k}=\mathbf{k}_r} + \mathrm{i}\mathbf{k}_i \cdot \left[\frac{\partial}{\partial \mathbf{k}} \overleftrightarrow{\mathbf{K}}(\omega, \mathbf{k}) \right]_{\omega=\omega_r, \mathbf{k}=\mathbf{k}_r} \tag{7.84}$$

while the Hermitian part remains as before. There is now a term involving $\mathbf{k}_i$. With the incorporation of this new term, Eq. (7.75) becomes

$$\left\langle \mathbf{E} \cdot \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \right\rangle = \left[\begin{array}{l} \dfrac{2\varepsilon_0 \omega_i}{4} \tilde{\mathbf{E}}^* \cdot \left[\dfrac{\partial}{\partial \omega} \omega \overleftrightarrow{\mathbf{K}}(\omega) \right]_{\omega=\omega_r} \cdot \tilde{\mathbf{E}} \\ + \dfrac{\varepsilon_0}{4} \tilde{\mathbf{E}}^* \cdot \left[2\omega \mathbf{k}_i \cdot \dfrac{\partial}{\partial \mathbf{k}} \overleftrightarrow{\mathbf{K}}(\omega, \mathbf{k}) \right]_{\substack{\omega=\omega_r, \\ \mathbf{k}=\mathbf{k}_r}} \cdot \tilde{\mathbf{E}} \end{array} \right] \mathrm{e}^{-2\mathbf{k}_i \cdot \mathbf{x} + 2\omega_i t}, \tag{7.85}$$

where we have explicitly written the exponential space-dependent factor $\exp(-2\mathbf{k}_i \cdot \mathbf{x})$.

What is the meaning of this new term involving $\mathbf{k}_i$? The answer to this question may be found by examining the Poynting flux for the situation where $\mathbf{k}_i$ is finite. Using the product rule and allowing for finite $\mathbf{k}_i$ shows

$$
\begin{aligned}
\langle \nabla \cdot \mathbf{P} \rangle &= \langle \nabla \cdot (\mathbf{E} \times \mathbf{H}) \rangle \\
&= \frac{1}{4} \nabla \cdot \left[\left(\tilde{\mathbf{E}}^* \times \tilde{\mathbf{H}} + \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right) \mathrm{e}^{-2\mathbf{k}_i \cdot \mathbf{x} + 2\omega_i t} \right] \\
&= \frac{1}{4} \left(-2\mathbf{k}_i \cdot \left(\tilde{\mathbf{E}}^* \times \tilde{\mathbf{H}} + \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right) \right) \mathrm{e}^{-2\mathbf{k}_i \cdot \mathbf{x} + 2\omega_i t}.
\end{aligned}
\tag{7.86}
$$

Comparison of Eqs. (7.85) and (7.86) shows that the factor $-2\mathbf{k}_i$ corresponds to the divergence operator acting on the product of two oscillating quantities and so the second term in Eq. (7.85) represents an *energy flux*. Since the Poynting vector $\mathbf{P}$ is the energy flux associated with the electromagnetic field, this additional energy flux must be identified as the energy flux associated with particle motion. Defining this flux as $\mathbf{T}$ it is seen that

$$
T_j = -\frac{\omega \varepsilon_0}{4} \tilde{\mathbf{E}}^* \cdot \left[\frac{\partial}{\partial k_j} \overleftrightarrow{\mathbf{K}} (\omega, \mathbf{k}) \right] \cdot \tilde{\mathbf{E}}
\tag{7.87}
$$

in the limit $\mathbf{k}_i \to 0$. For small but finite $\mathbf{k}_i$, ω_i the generalized Poynting theorem can be written as

$$
-2\mathbf{k}_i \cdot (\mathbf{P} + \mathbf{T}) + 2\omega_i (w_E + w_B + \bar{w}_{part}) = 0.
\tag{7.88}
$$

We now define the generalized group velocity $\mathbf{v}_g$ to be the velocity with which wave energy is transported. This velocity is the total energy flux divided by the total energy density, i.e.,

$$
\mathbf{v}_g = \frac{\mathbf{P} + \mathbf{T}}{w_E + w_B + \bar{w}_{part}}.
\tag{7.89}
$$

The bar in $\bar{w}_{part}$ indicates that this term is the *difference* between the particle energy with the wave and the particle energy without the wave and so could, in principle, be negative.

7.7 Negative energy waves

A curious consequence of this analysis is that a wave can have a *negative* energy density. The field energy densities w_E and w_B are positive-definite, but the particle energy density $\bar{w}_{part}$ can have either sign and in certain circumstances can be sufficiently negative to make the total wave-energy density negative. This surprising possibility can occur because the wave-energy density defined in Eq. (7.78) is actually the *change* in the total system energy density on going from a situation

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where there is no wave to a situation where there is a wave. Typically, negative energy waves occur when the equilibrium has a steady-state flow velocity and there exists a mode that reduces the average kinetic energy of the particles to a value below the initial equilibrium value. The growth of this type of mode taps the free energy in the flow.

As an example of a negative energy wave, we consider the situation where unmagnetized cold electrons stream with velocity v_0 through a background of infinitely massive ions. As shown in Section 5.2 the electrostatic dispersion for this simple 1-D situation with flow involves a parallel dielectric having a Doppler-shifted frequency, i.e., the dispersion relation is

$$P(\omega, k) = 1 - \frac{\omega_{pe}^2}{(\omega - kv_0)^2} = 0; \tag{7.90}$$

here it has been assumed that there exists a neutralizing background of infinitely massive ions. Since the plasma is unmagnetized, its dielectric tensor is simply $\overleftrightarrow{\mathbf{K}}(\omega, \mathbf{k}) = P(\omega, k)\overleftrightarrow{\mathbf{I}}$. Using Eq. (7.79), the wave-energy density is

$$w = \frac{\varepsilon_0|\tilde{E}|^2}{4}\frac{\partial}{\partial\omega}(\omega P(\omega, k)) = \frac{\varepsilon_0|\tilde{E}|^2}{2}\frac{\omega\omega_{pe}^2}{(\omega - kv_0)^3}. \tag{7.91}$$

However, the dispersion relation, Eq. (7.90), shows that

$$\omega = kv_0 \pm \omega_{pe} \tag{7.92}$$

so Eq. (7.91) can be recast as

$$w = \frac{\varepsilon_0|\tilde{E}|^2}{2}\left(1 \pm \frac{kv_0}{\omega_{pe}}\right). \tag{7.93}$$

Thus, if $kv_0 > \omega_{pe}$ and the minus sign is selected, the wave has *negative* energy density.

This result can be verified by direct calculation of the change in system energy density due to growth of the wave. When there is no wave, the electric field is zero and the system energy density w^{sys} is simply the beam kinetic energy density

$$w_0^{sys} = \frac{1}{2}n_0 m_e v_0^2. \tag{7.94}$$

Now consider a one-dimensional electrostatic wave with electric field $E = \mathrm{Re}\left(\tilde{E}\mathrm{e}^{\mathrm{i}kx-\mathrm{i}\omega t}\right)$. The average energy density of the system with this wave is

$$w_{wave}^{sys} = \frac{\varepsilon_0|\tilde{E}\;|^2}{4} + \left\langle\frac{1}{2}n(x, t)m_e v(x, t)^2\right\rangle \tag{7.95}$$

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so the change in system energy density due to the wave is

$$\bar{w}^{sys} = w^{sys}_{wave} - w^{sys}_0 = \frac{\varepsilon_0 |\tilde{E}\ |^2}{4} + \left\langle \frac{1}{2} \left[n_0 + n_1(x,t)\right] m_e \left[v_0 + v_1(x,t)\right]^2 \right\rangle - \frac{1}{2} n_0 m_e v_0^2, \tag{7.96}$$

where

$$n_1(x,t) = \mathrm{Re}\left(\tilde{n}\mathrm{e}^{\mathrm{i}kx - \mathrm{i}\omega t}\right), \quad v_1(x,t) = \mathrm{Re}\left(\tilde{v}\mathrm{e}^{\mathrm{i}kx - \mathrm{i}\omega t}\right). \tag{7.97}$$

Since odd powers of oscillating quantities vanish upon time averaging, Eq. (7.96) reduces to

$$\begin{aligned} \bar{w}^{sys} &= \frac{\varepsilon_0 |\tilde{E}|^2}{4} + n_0 m_e \left\langle \frac{1}{2} v_1^2 + \frac{n_1}{n_0} v_1 v_0 \right\rangle \\ &= \frac{\varepsilon_0 |\tilde{E}|^2}{4} + \frac{1}{2} n_0 m_e \left\langle v_1^2 \right\rangle + m_e v_0 \left\langle n_1 v_1 \right\rangle . \end{aligned} \tag{7.98}$$

The linearized continuity equation gives

$$-\mathrm{i}(\omega - k v_0)\tilde{n} + n_0 \mathrm{i} k \tilde{v} = 0 \tag{7.99}$$

or

$$\frac{\tilde{n}}{n_0} = \frac{k\tilde{v}}{\omega - k v_0}. \tag{7.100}$$

The electron quiver velocity in the wave is

$$\tilde{v} = \frac{q\tilde{E}}{-\mathrm{i}(\omega - k v_0) m_e} \tag{7.101}$$

so

$$\left\langle v_1^2 \right\rangle = \frac{1}{2} \frac{q^2 |\tilde{E}|^2}{(\omega - k v_0)^2 m_e^2} \tag{7.102}$$

and

$$\left\langle n_1 v_1 \right\rangle = \frac{k}{2} \frac{n_0 q^2 |\tilde{E}|^2}{(\omega - k v_0)^3 m_e^2}. \tag{7.103}$$

We may now evaluate Eq. (7.98) to obtain

$$\begin{aligned} \bar{w}^{sys} &= \frac{\varepsilon_0 |\tilde{E}|^2}{4} + n_0 m_e \left[\frac{q^2 |\tilde{E}|^2}{4(\omega - k v_0)^2 m_e^2} + \frac{k v_0}{2} \frac{q^2 |\tilde{E}|^2}{(\omega - k v_0)^3 m_e^2} \right] \\ &= \frac{\varepsilon_0 |\tilde{E}|^2}{4} \left[1 + \frac{\omega_{pe}^2}{(\omega - k v_0)^2} + \frac{2\omega_{pe}^2 k v_0}{(\omega - k v_0)^3} \right] \\ &= \frac{\varepsilon_0 |\tilde{E}|^2}{2} \left[1 \pm \frac{k v_0}{\omega_{pe}} \right], \end{aligned} \tag{7.104}$$

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where Eq. (7.92) has been repeatedly invoked. This is the same as Eq. (7.93). The energy flux associated with this wave is also negative (cf. assignments). However, the group velocity is positive (cf. assignments) because the group velocity is the ratio of a negative energy flux to a negative energy density.

Dissipation affects negative energy waves in a manner opposite to its effect on positive energy waves. This can be seen by Taylor expanding the dispersion relation $P(\omega, k) = 0$ as done in Eq. (5.87) to obtain

$$\omega_i = -\frac{P_i}{[\partial P_r/\partial\omega]_{\omega=\omega_r}}. \tag{7.105}$$

Expanding Eq. (7.91) gives

$$\bar{w} = \frac{\varepsilon_0 \omega |E|^2}{4} \frac{\partial P}{\partial \omega} \tag{7.106}$$

so a negative energy wave has $\omega \partial P/\partial\omega < 0$. If the dissipative term P_i is the same for both positive and negative waves, then, for a given sign of ω, the critical difference between positive and negative waves lies in the sign of $\partial P/\partial\omega$. Equation (7.105) shows that ω_i has opposite signs for positive and negative energy waves. This means that, in contrast to positive energy waves, dissipation *destabilizes* negative energy waves. This is because dissipation enables the tapping of available free energy by the negative energy wave. Since any real system generally has some dissipation, if a negative energy wave is an allowed mode in a given real system, the negative energy wave will spontaneously develop and grow at the expense of the free energy in the system (e.g., the free energy in the streaming particles).

7.8 Assignments

1. Consider the problem of short-wave radio transmission. Let x be the horizontal direction and z be the vertical direction. A short-wave radio antenna is designed to radiate most of the transmitter power into a specified k_x and k_z by careful control over the Fourier transform of the antenna geometry.
   (i) What is the frequency range of short-wave radio communications?
   (ii) For ionospheric parameters and the majority of the short-wave band, should ionospheric plasma be considered as magnetized or unmagnetized?
   (iii) What is the appropriate dispersion for short-wave radio waves (hint: it is very simple)?
   (iv) Using Snell's law and geometric optics, sketch the trajectory of a radio wave propagating from a terrestrial antenna (hint: consider k_x and k_z radiated by the antenna and consider the group velocity trajectory and also what happens when $\omega = \omega_{pe}(z)$).
2. Using geometric optics discuss qualitatively with sketches how a low-frequency wave could act as a lens for a high-frequency wave.

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3. Using the example of an electrostatic electron plasma wave (dispersion $\omega^2 = \omega_{pe}^2(1 + 3k^2\lambda_{de}^2)$) show that the generalized group velocity as defined in Eq. (7.89) gives the same group velocity as found by direct evaluation of $\partial\omega/\partial\mathbf{k}$.
4. Calculate the wave-energy density, wave-energy flux, and group velocity for the electrostatic wave that can occur when a beam of cold electrons streams with velocity $\mathbf{v}_0$ through a neutralizing background of infinitely massive ions. Discuss the signs of these three quantities.

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